PE- Introduction

Definition

- Application of solid state electronics for the control and conversion of electric power.
- Power Electronics- an oxymoron!

PE Switches

The success of any technology lies on three factors:-

(1) It should be highly efficient.

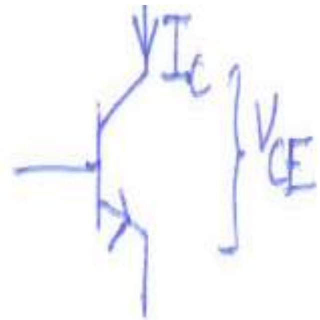
(2) It should be reliable.

(3) It should have low weight, cost & Size.

* High η ⇒ Low loss ⇒ Lesser cooling requirement ⇒ Packaging can be done tightly ⇒ Size comes down.

Why PE equipments are efficient?

Consider



Power loss $(P_L) = V_{CE} I_C$.

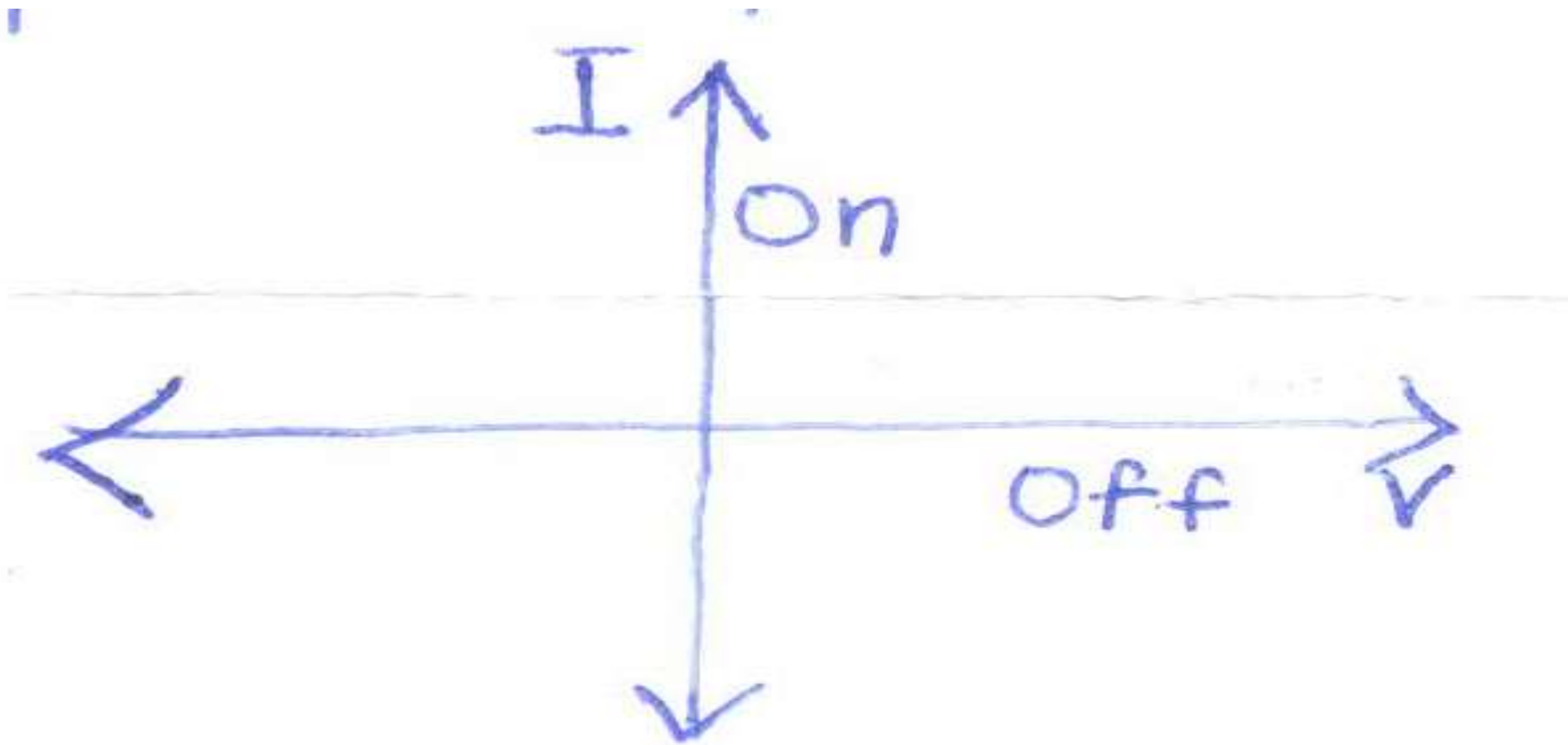
In Active region $V_{CE} \uparrow \Rightarrow P_L \uparrow$. In Saturation region $V_{CE} \downarrow \Rightarrow P_L \downarrow$.

* In PE equipments L & C & SC devices operated in saturation / cutoff region are used $\therefore$ higher η.

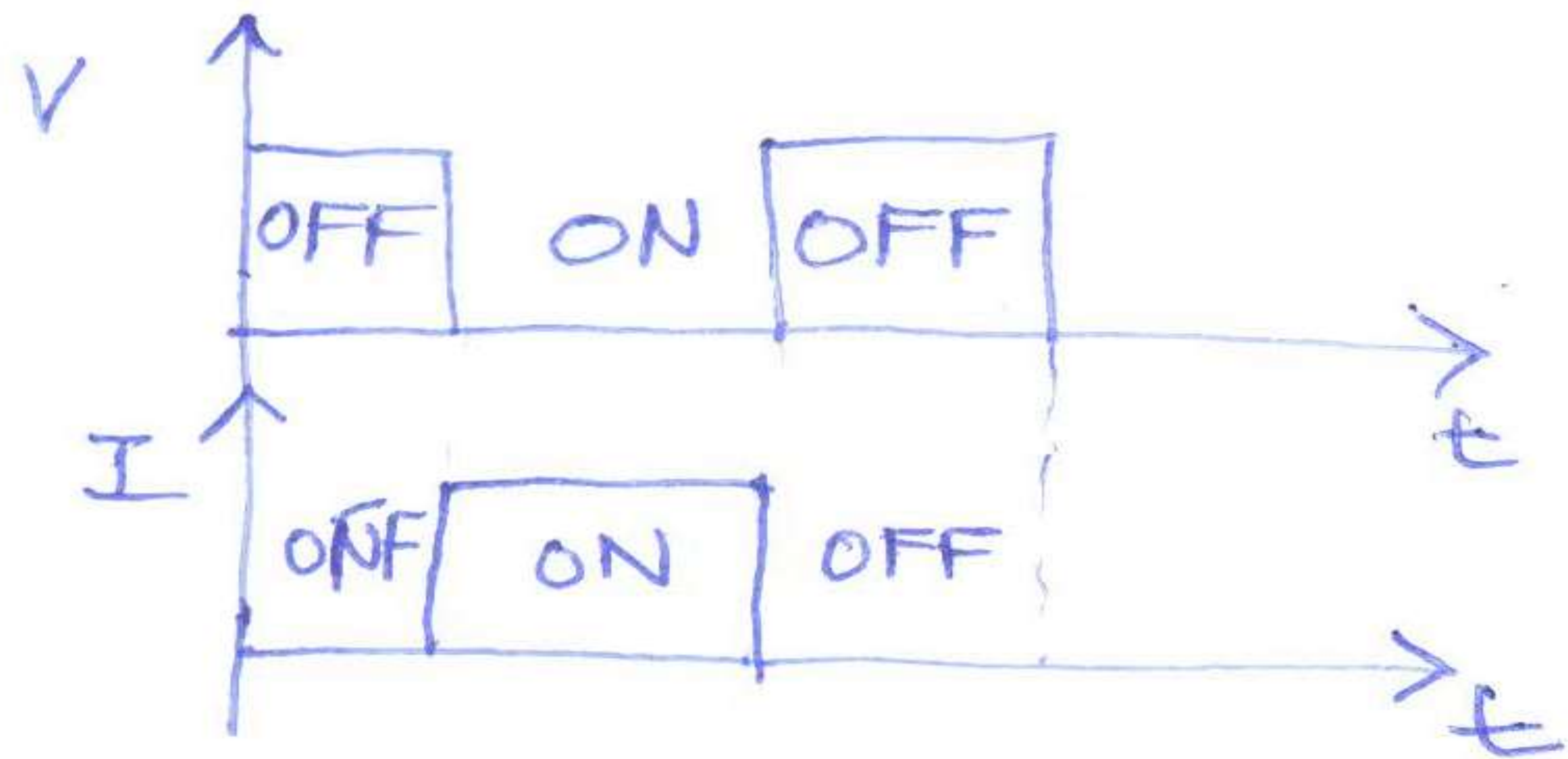
Ideal Switch- Properties

- Power dissipated is zero i.e conduction, blocking and switching losses are zero.

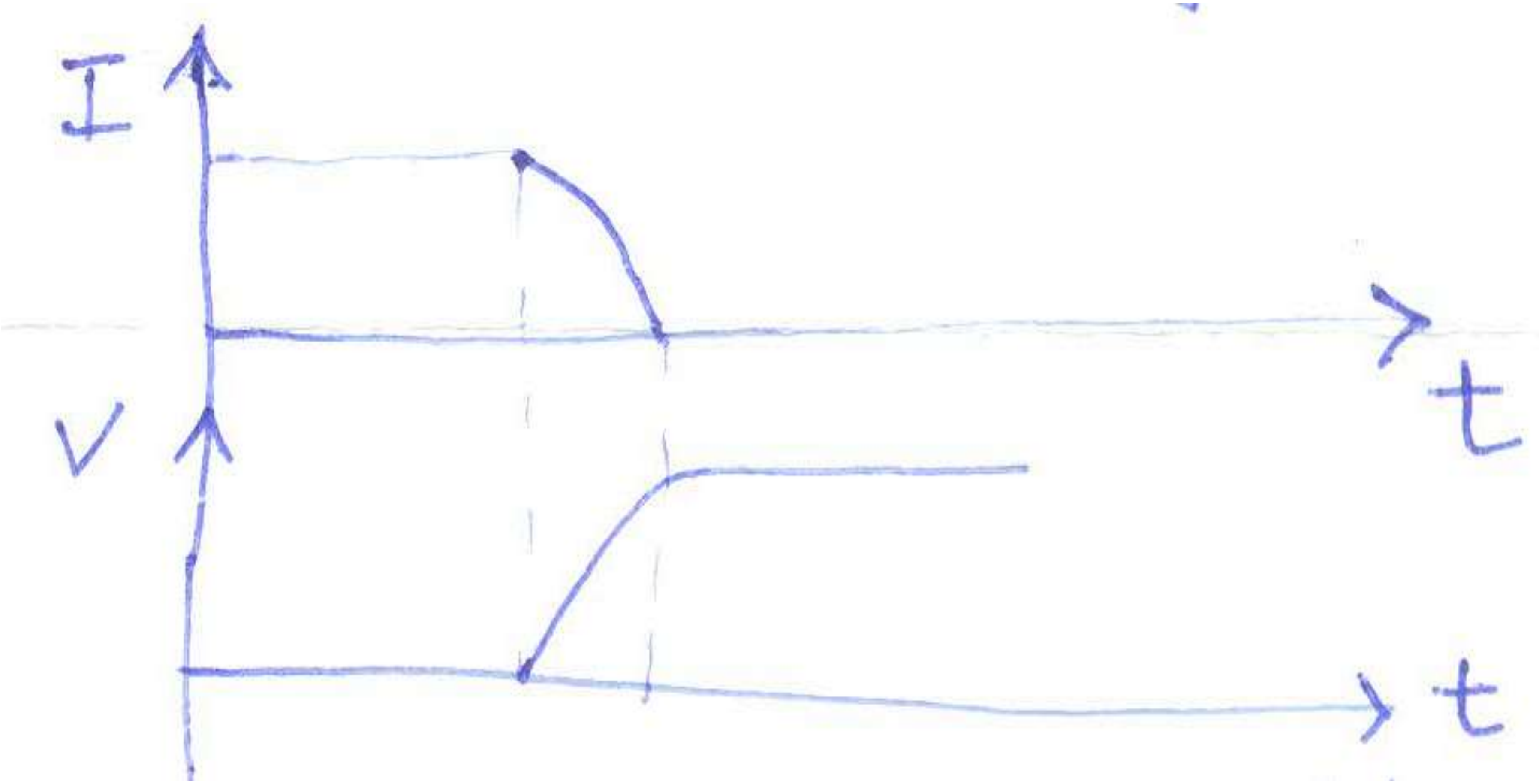
Ideal Switch VI Characteristics



Switching Characteristics of Ideal Switch



Switching Characteristics of Practical Switch



Classification of Power Electronic Switches

- Uncontrolled- ON/ OFF determined by state of power circuit.
- Semi-controlled- One state is controllable via control terminal.
- Fully-controlled- ON/ OFF determined by control terminal.

Characteristics & Switching Behaviour of Power Electronic Switches

Uncontrolled Switch

E.g. Diode and DIAC

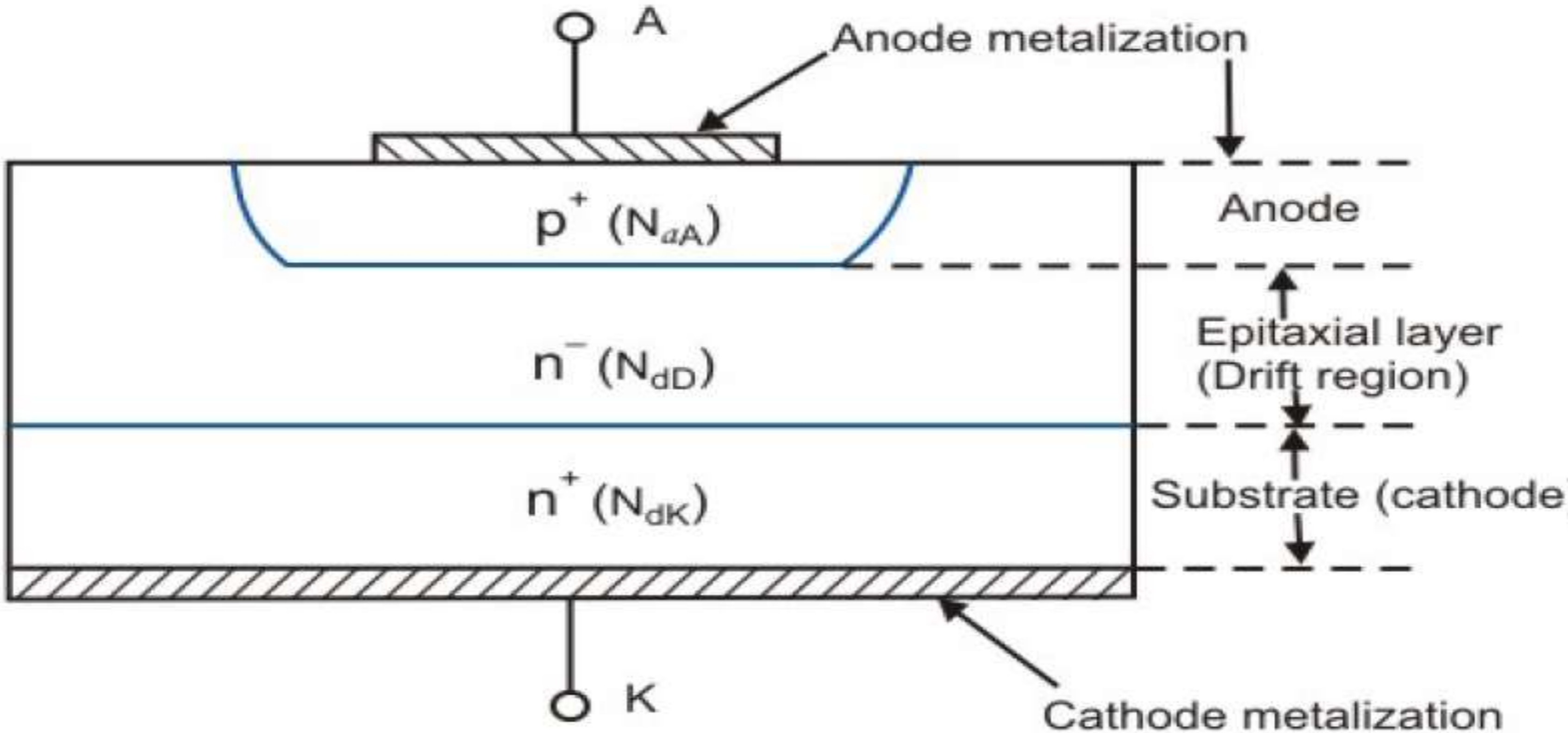
POWER DIODE

- Power semiconductor diode is the “power level” counter part of the “low power signal diodes.”
- These power devices are required to carry up to several KA of current under forward bias condition and block up to several KV under reverse biased condition.
- These extreme requirements call for important structural changes in a power diode.
- These structural modifications are generic in the sense that the same basic modifications are applied to all other low power semiconductor devices to scale up their power capabilities.

Construction and Characteristics of Power Diodes

- Power Diodes of largest power rating are required to conduct several kilo amps of current in the forward direction with very little power loss while blocking several kilo volts in the reverse direction.
- Large blocking voltage requires wide depletion layer in order to restrict the maximum electric field strength below the “impact ionization” level.
- Space charge density in the depletion layer should also be low in order to yield a wide depletion layer for a given maximum electric field strength.
- These two requirements will be satisfied in a lightly doped p-n junction diode of sufficient width to accommodate the required depletion layer.

- Such a construction, however, will result in a device with high resistivity in the forward direction.
- Consequently, the power loss at the required rated current will be unacceptably high.
- On the other hand if forward resistance is reduced by increasing the doping level, reverse break down voltage will reduce.
- This apparent contradiction in the requirements of a power diode is resolved by introducing a lightly doped “drift layer” of required thickness between two heavily doped p and n layers.



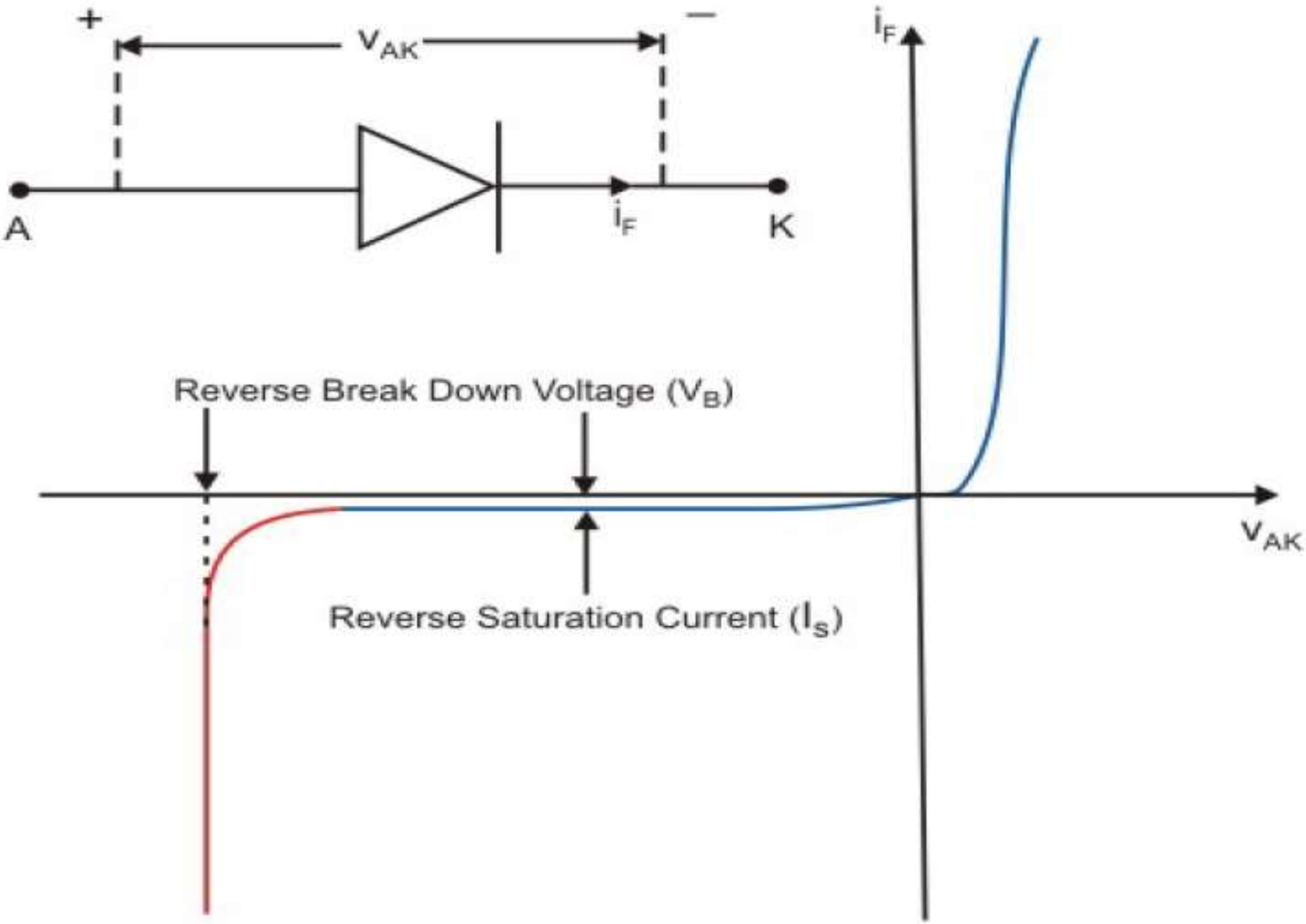
(c)

(a)

Diagram of a power diode (c) schematic cross section (a) circuit symbol.

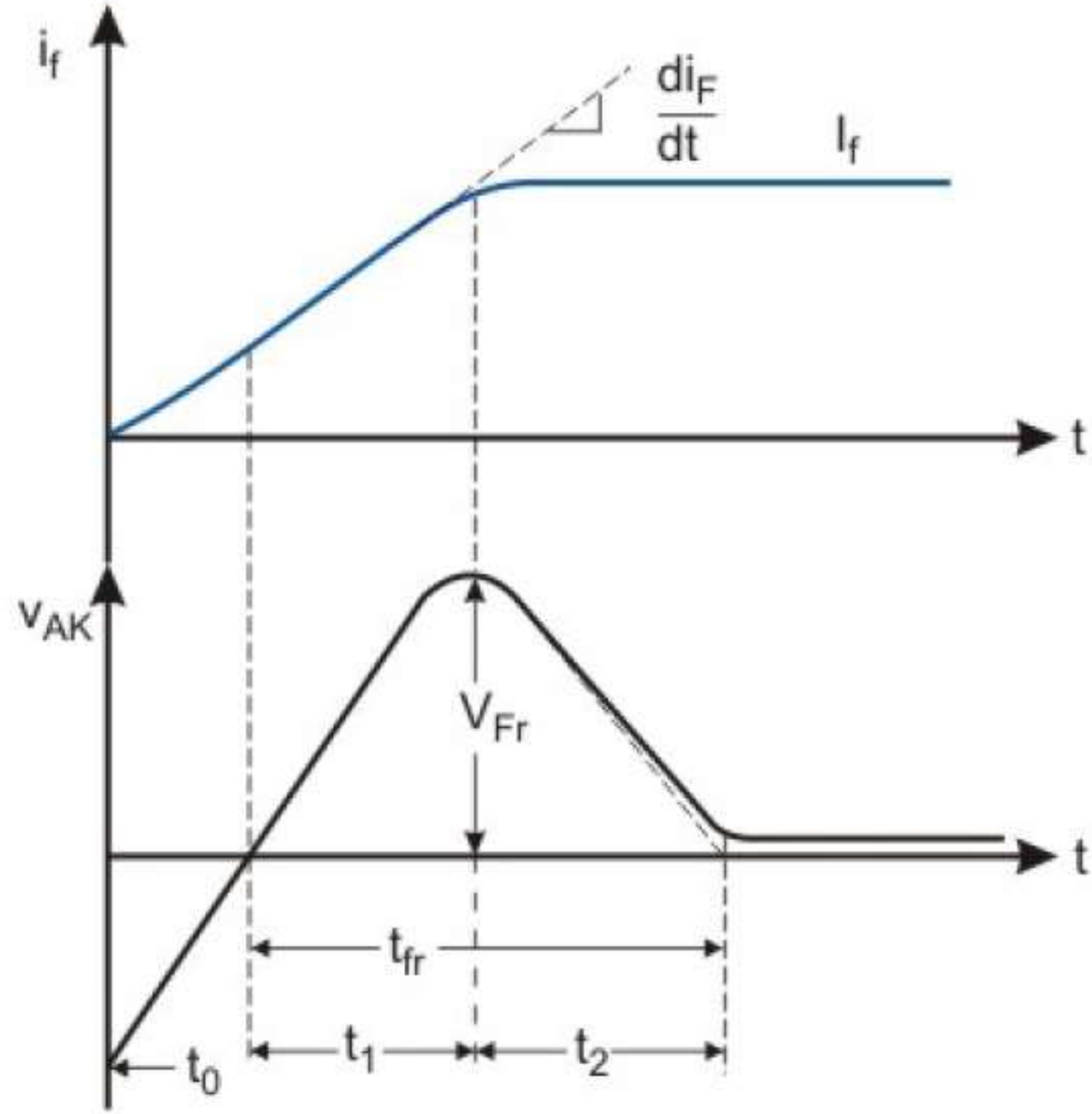
Can a lightly doped p layer be used as a drift layer?

Mobility is a trade off between number of carriers and collision.



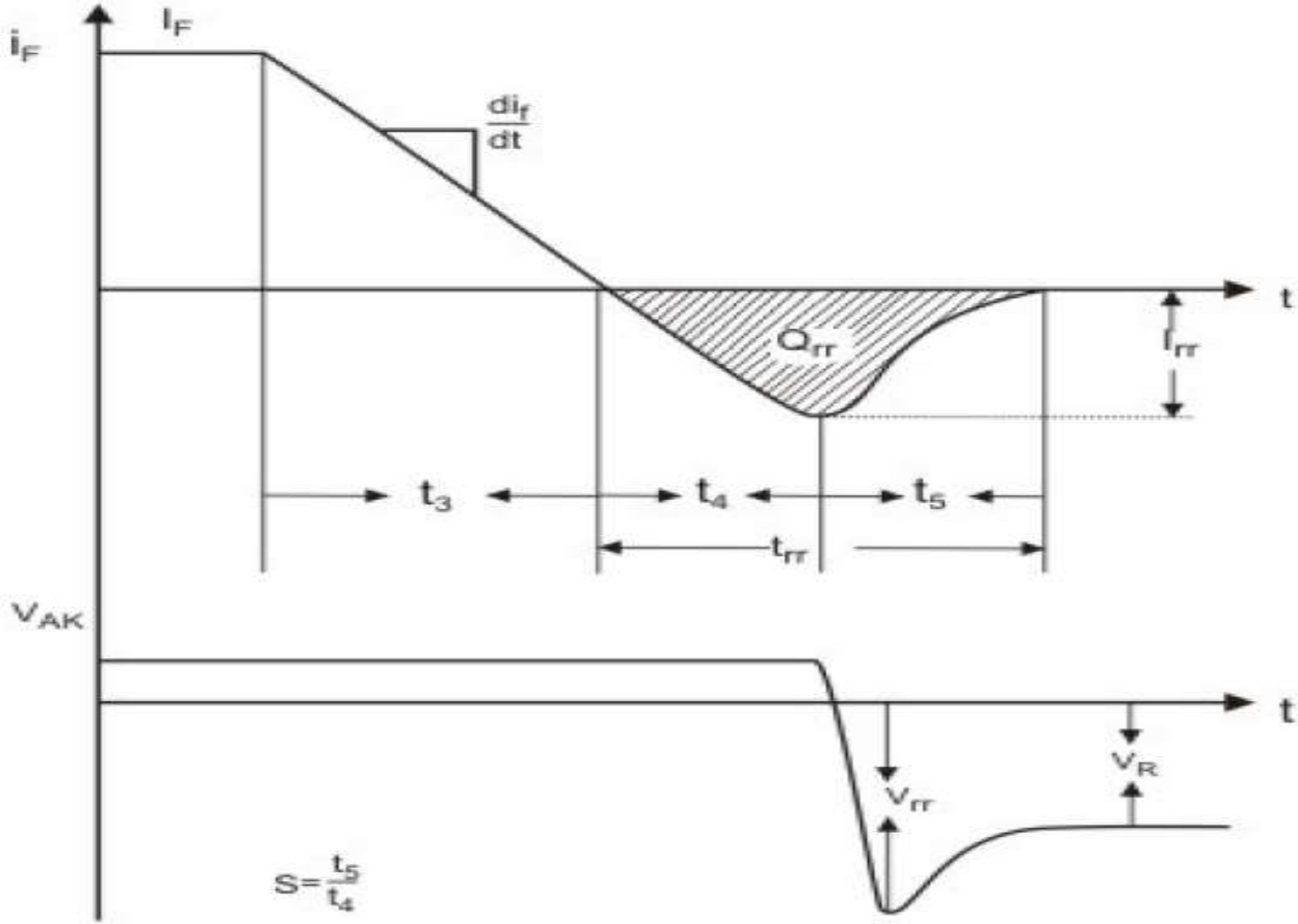
Volt-Ampere (i-v) characteristics of a p-n junction diode

Diode Turn On Characteristics



Forward current and voltage waveforms of a power diode during Turn On operation.

Diode Turn Off Characteristics

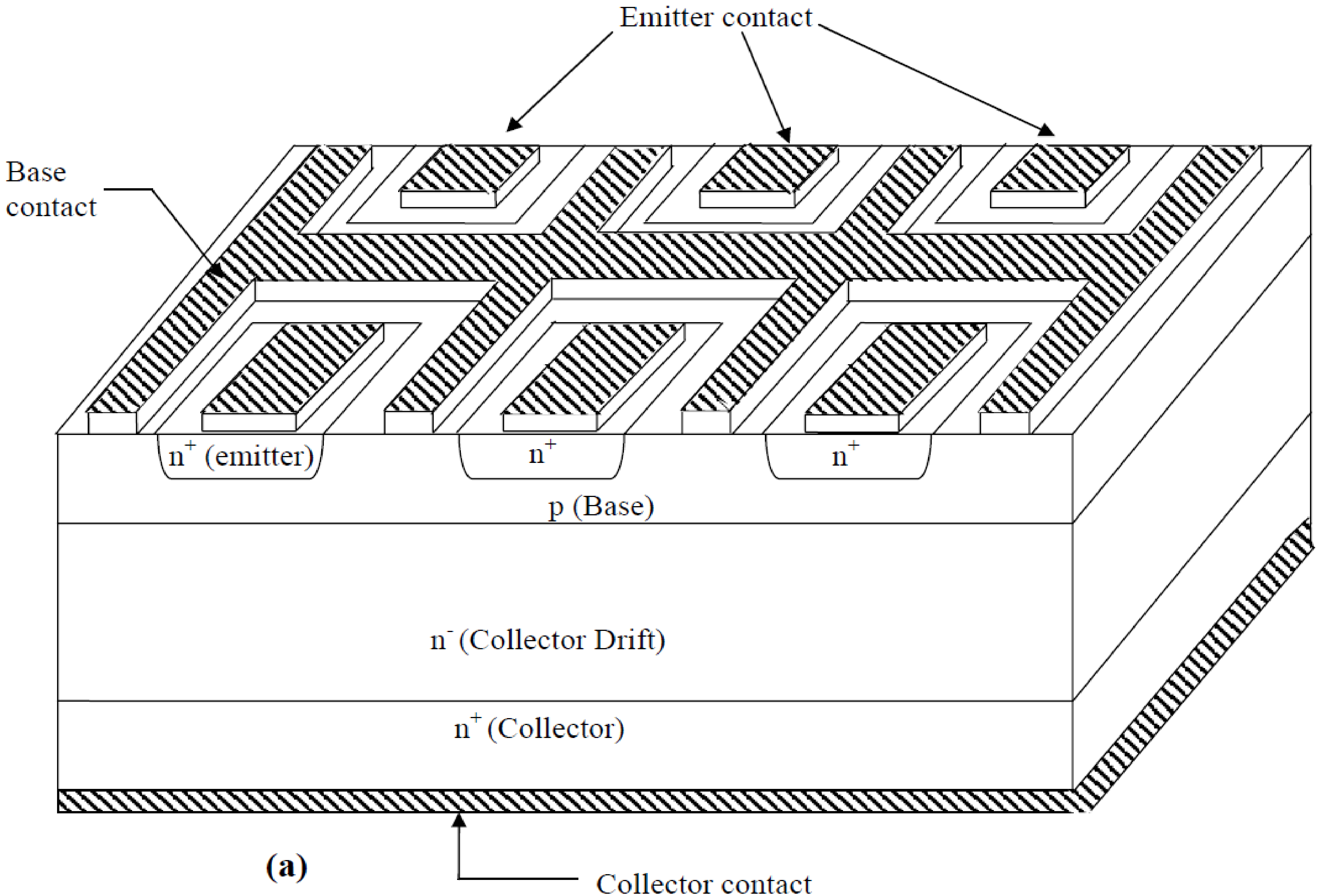


Reverse Recovery characteristics of a power diode

Fully Controlled Switch

E.g. GTO, MOSFET, IGBT, MCT & Power BJT

Power BJT

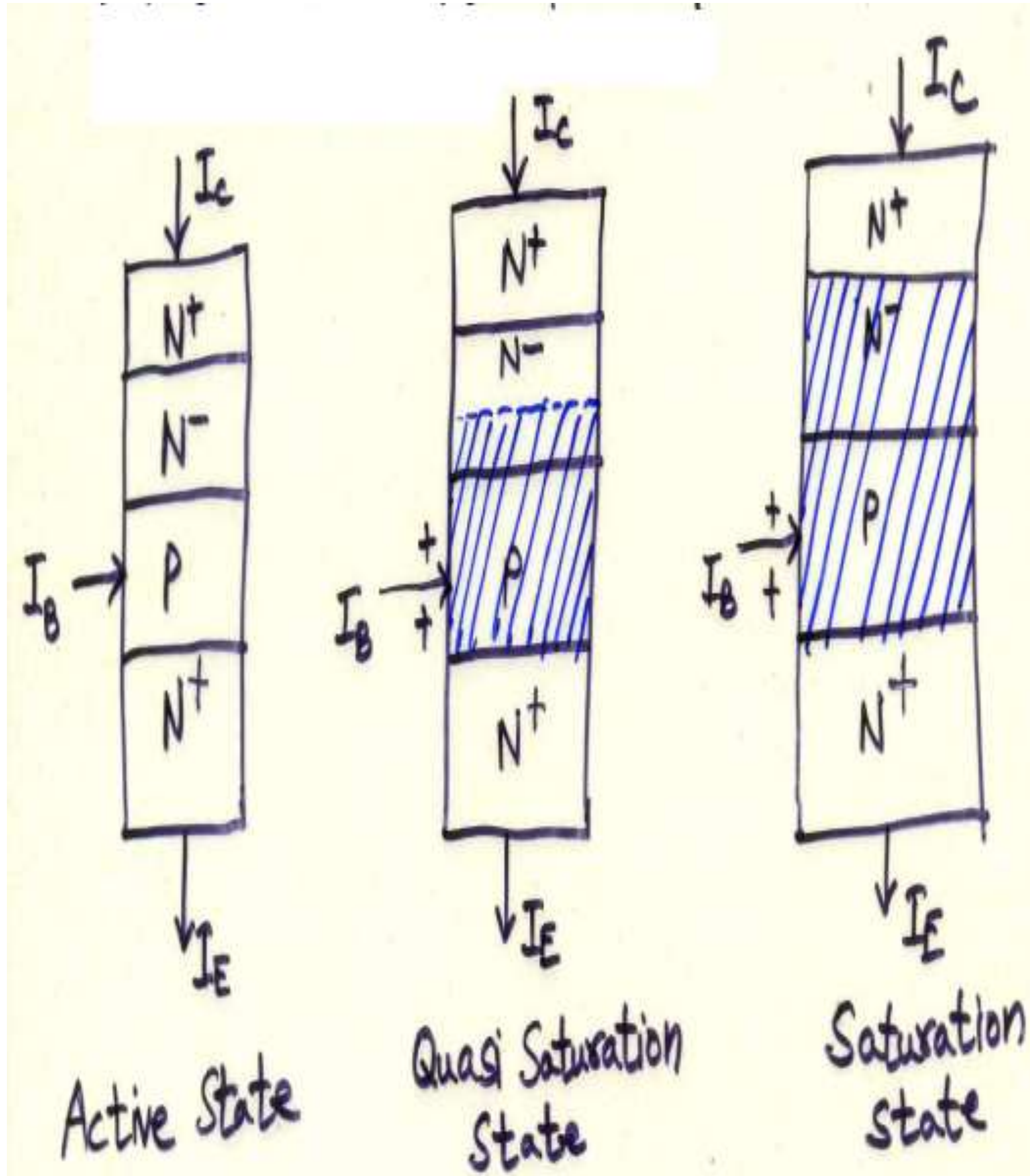


Emitter contact
Base contact
n+ (emitter)
n+
n+
p (Base)
n- (Collector Drift)
n+ (Collector)
Collector contact

(a)

Basics

1) PBJT is operated at the edge of saturation in the ON state.
2) PBJT is a current controlled **minority carrier** device.
3) Minority carrier device criteria?
4) Minority carrier devices undergo second break down.



Second Breakdown

- The collector voltage drop is often accompanied by significant rise in the collector current and a substantial increase in the power dissipation.
- This dissipation is not uniformly spread over the entire volume of the device but is concentrated in highly localized regions.
- This localized heating is a combined effect of the intrinsic non uniformity of the collector current density distribution across the cross section of the device and the negative temperature coefficient of resistivity of minority carrier devices which leads to the formation of “current filaments” (localized areas of very high current density) by a positive feed-back mechanism.
- Once current filaments are formed localized “thermal runaway” quickly takes the junction temperature beyond the safe limit and the device is destroyed.

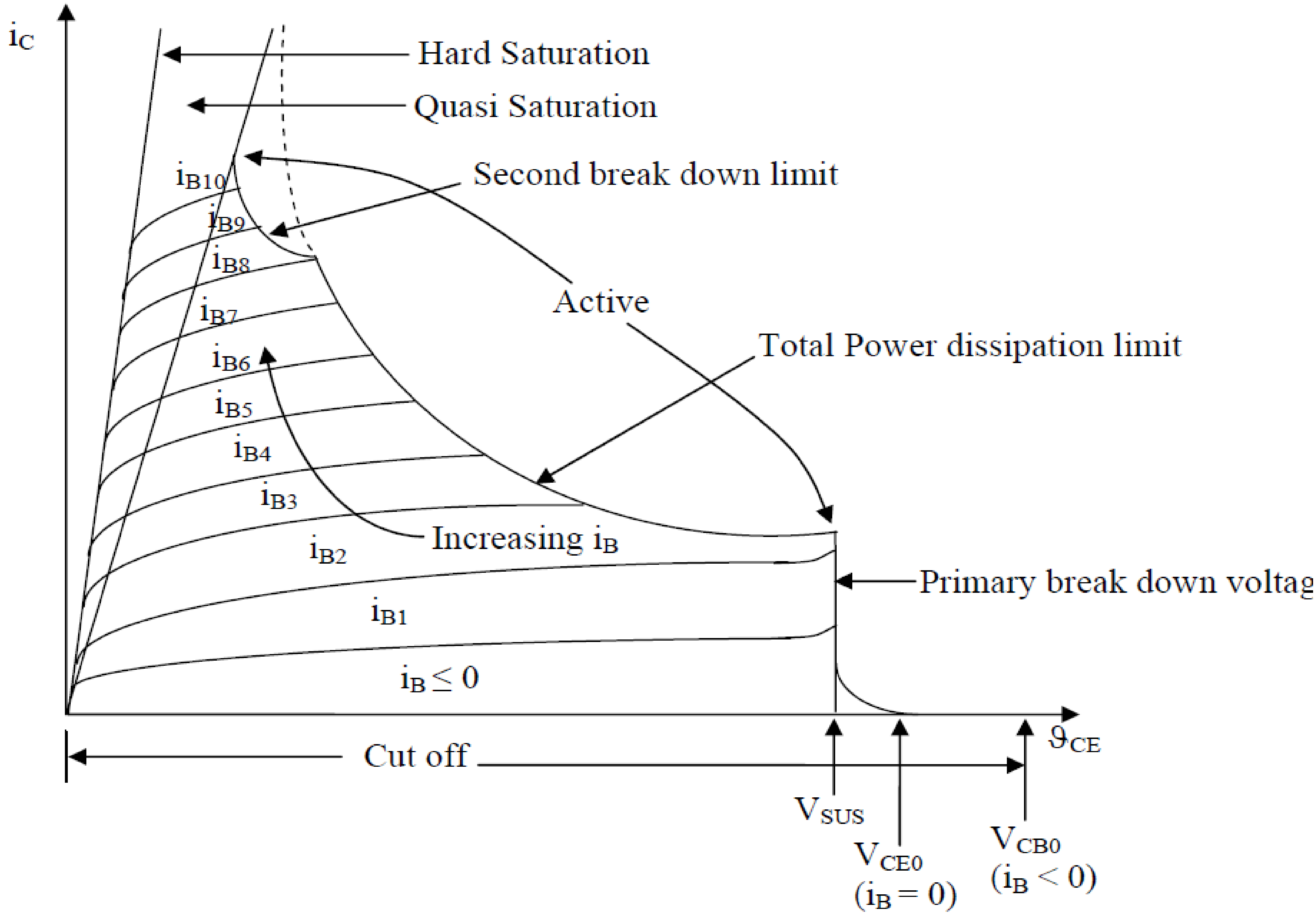
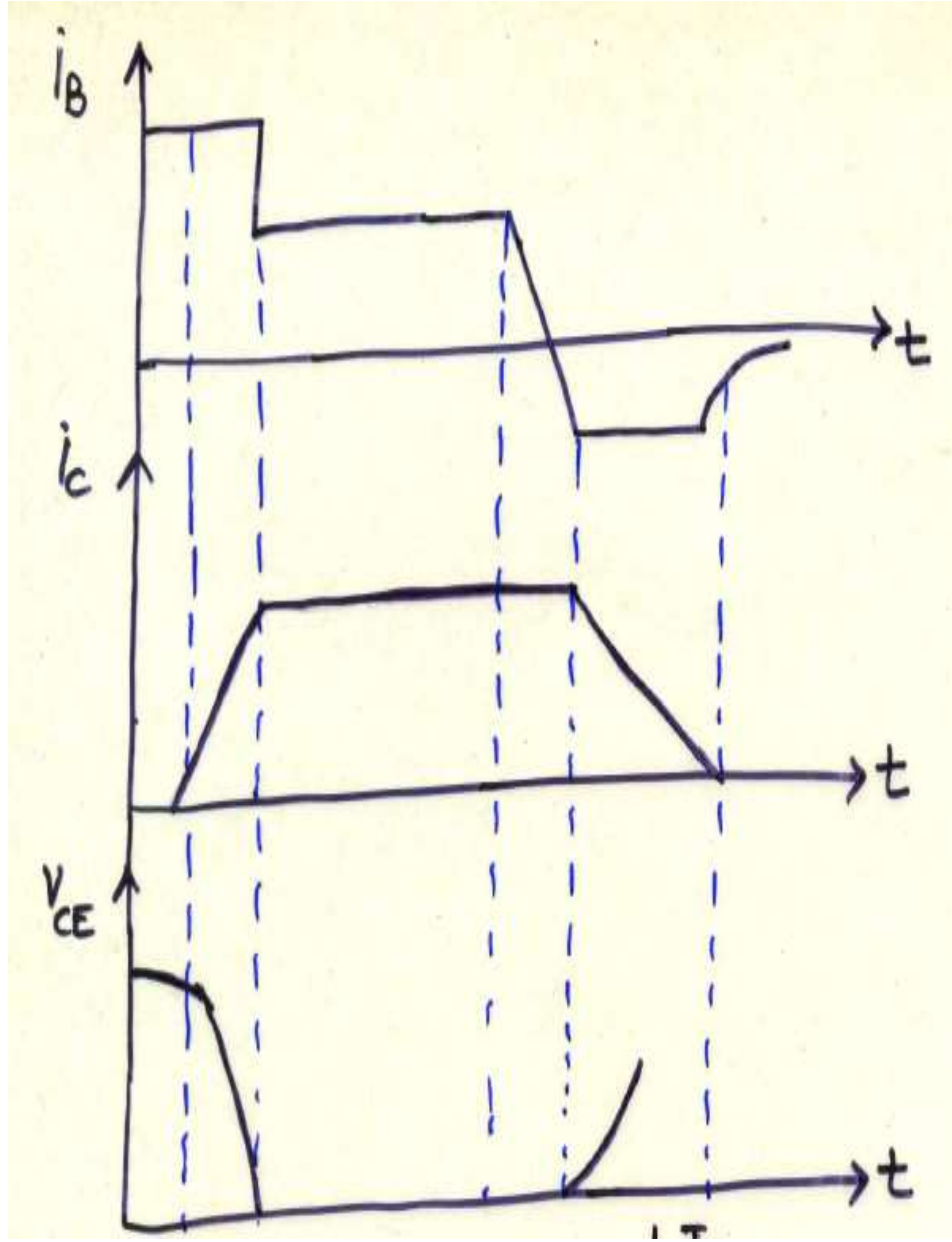


Fig. 3.4 Output ($i_c - v_{CE}$) characteristics of an n – p – n type Power Transistor

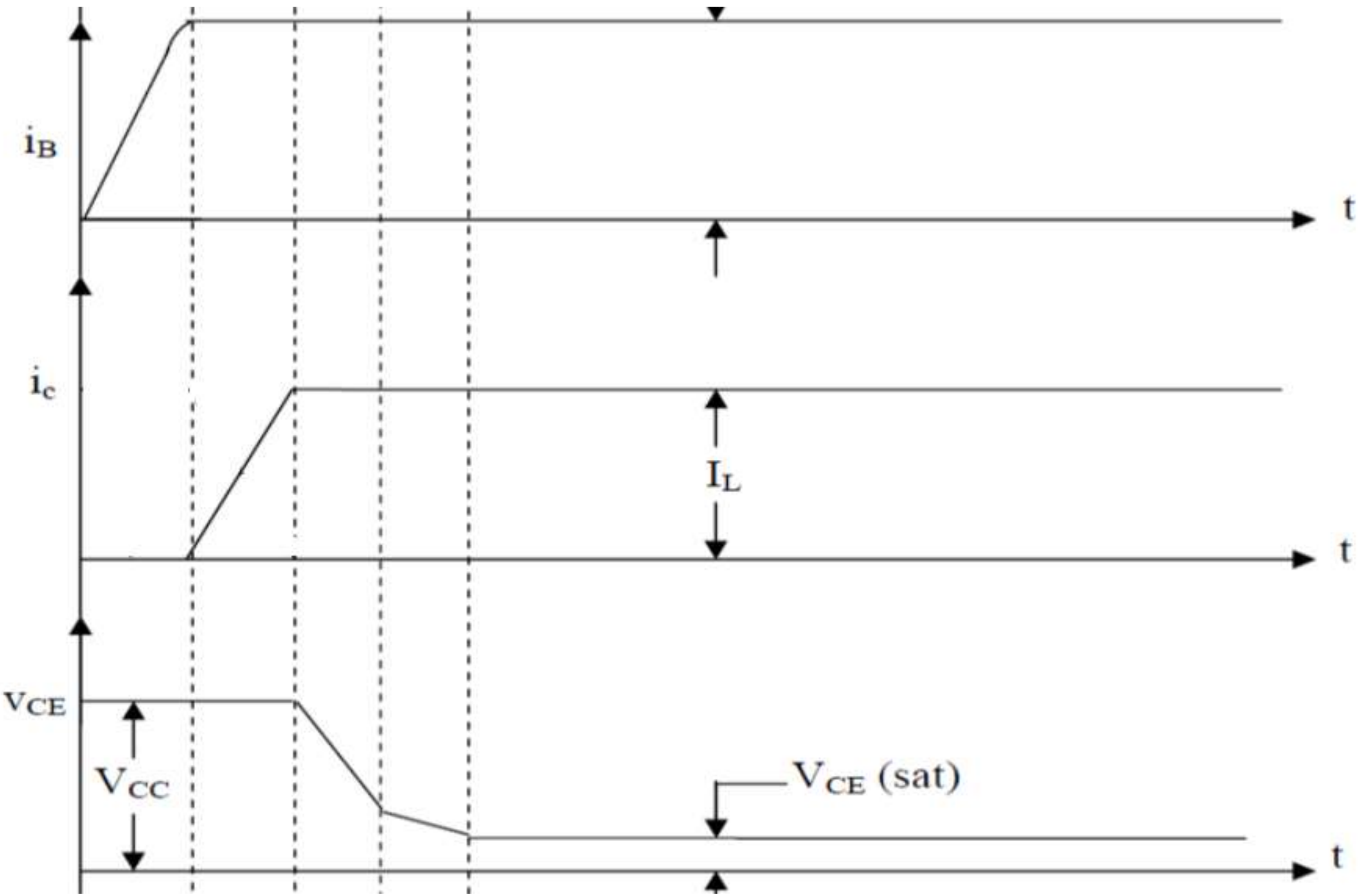
Switching Characteristics

1) During turn ON, there is a progressive accumulation of charges in the base, which increases collector current.
2) To reduce turn ON time, a higher value of base current is supplied initially.
3) Base current is reduced after turn ON to avoid letting the transistor to get into deep saturation.
4) Operating in quasi saturation helps decrease the turn OFF time.
5) Turn OFF is achieved by making base current negative instead of zero.

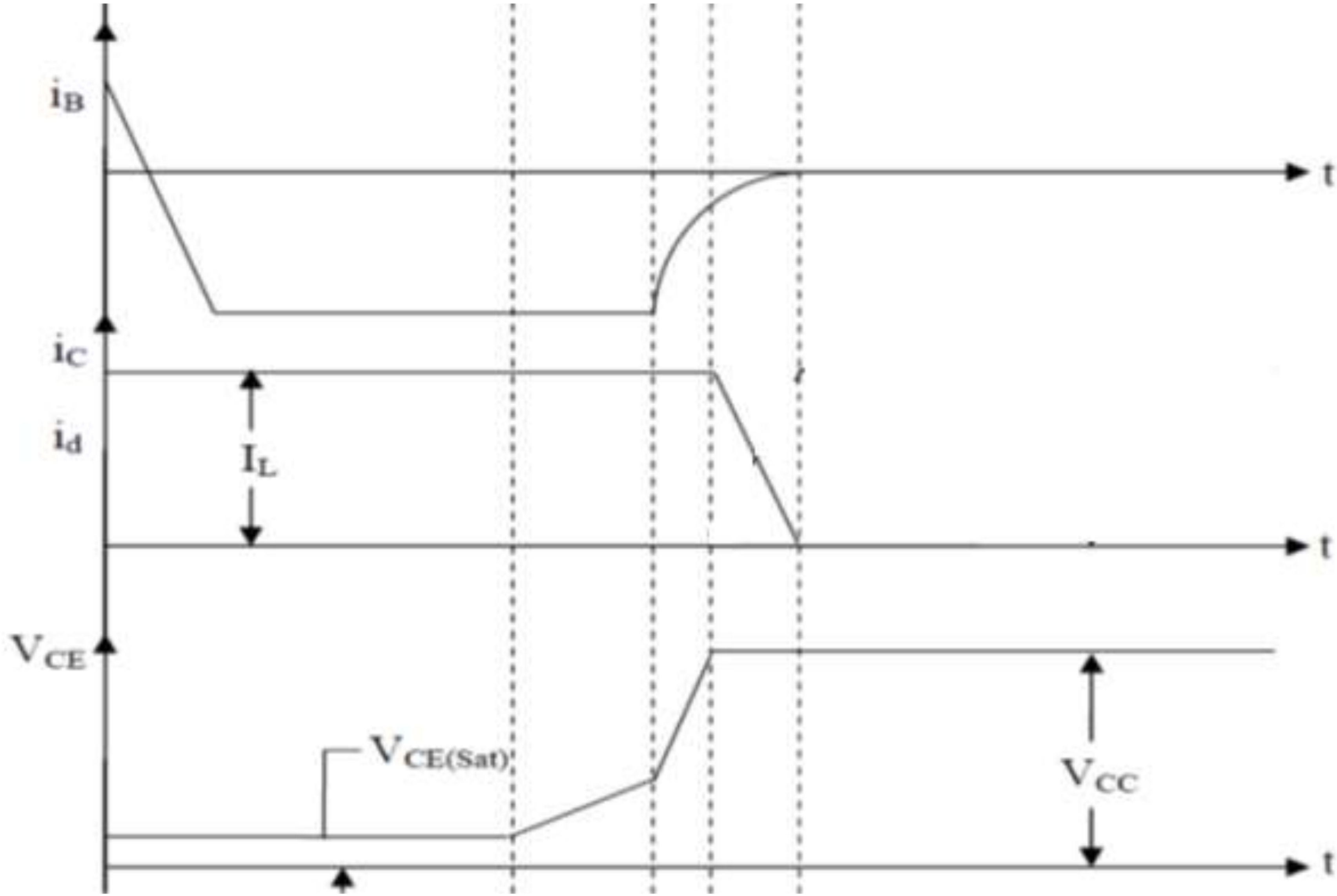


Switching characteristics= Turn ON + Turn Off Characteristics

PBJT Turn ON Characteristics



PBJT Turn OFF Characteristics



PBJT: NPN vs PNP (practical difference)

Semi Controlled Switch

E.g. Thyristor and TRIAC

Thyristor/ SCR

- P-N-P-N transistor switch concept introduced in 1956 by J.L. Moll and others at Bell Laboratories.
- Thyristor is a four layer, three terminal, minority carrier semi-controlled device.
- It can be turned on by a current signal but can not be turned off without interrupting the main current.
- It can block voltage in both directions but can conduct current only in one direction.

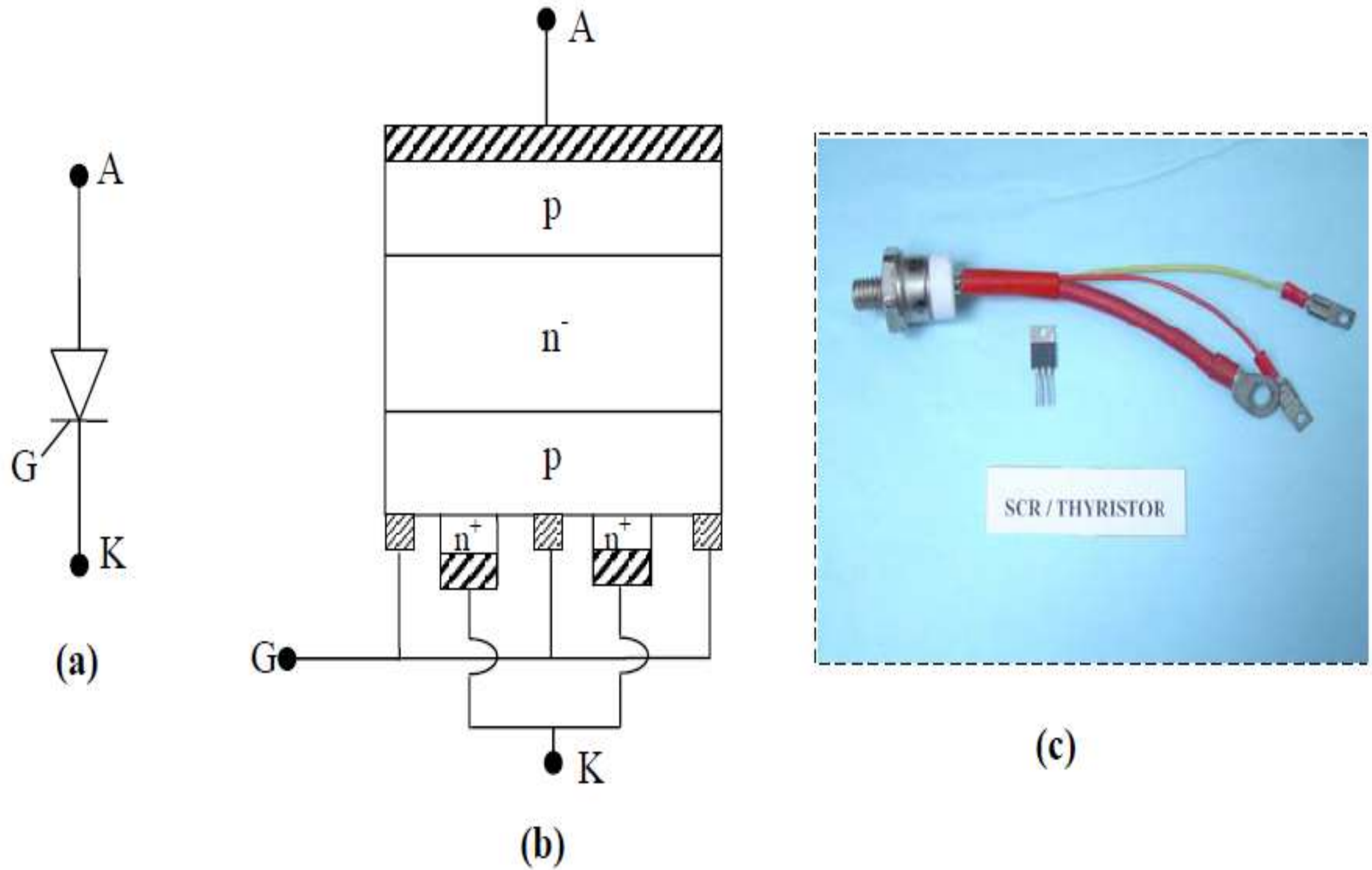


Fig. 4.1: Constructional features of a thysistor
(a) Circuit Symbol, (b) Schematic Construction, (c) Photograph

Modes of Operation (Role of J1, J2 and J3)

- Forward Blocking
- Reverse Blocking
- Forward Conduction

Basic Operating Principle

Two Transistor Analogy

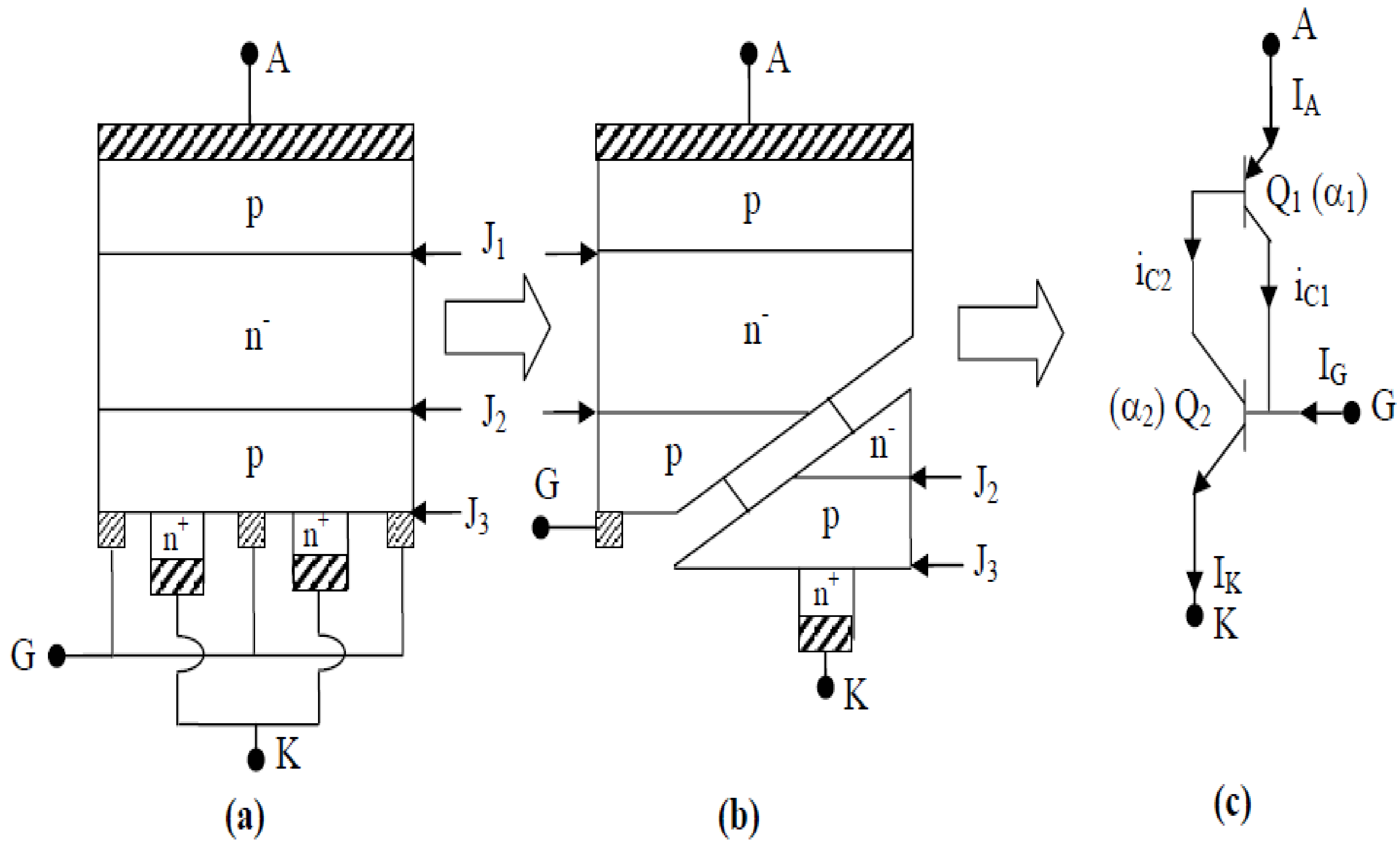


Fig. 4.2: Two transistor analogy of a thyristor construction.
(a) Schematic Construction, (b) Schematic division in component transistor
(c) Equivalent circuit in terms of two transistors.

$$i_{C1} = \alpha_1 I_A + I_{CO1} \quad \cdots\cdots (i)$$

$$i_{C2} = \alpha_2 I_K + I_{CO2} \quad \cdots\cdots (ii)$$

$$i_{C1} + i_{C2} = I_A \quad \cdots\cdots (iii)$$

$$I_A + I_G = I_K \quad \cdots\cdots (iv)$$

From (i) & (ii)

$$i_{C1} + i_{C2} = \alpha_1 I_A + \alpha_2 I_K + I_{CO1} + I_{CO2}$$

$$\Rightarrow I_A = \alpha_1 I_A + \alpha_2 I_K + I_{CO} \quad \text{[From (iii)]}$$

$$\Rightarrow I_A = \alpha_1 I_A + \alpha_2 (I_A + I_G) + I_{CO} \quad \text{[From (iv)]}$$

$$\Rightarrow I_A = \frac{\alpha_2 I_G + I_{CO}}{1 - (\alpha_1 + \alpha_2)}$$

Effect of Interchanging Transistor C and E terminals!

Amplification versus Switching

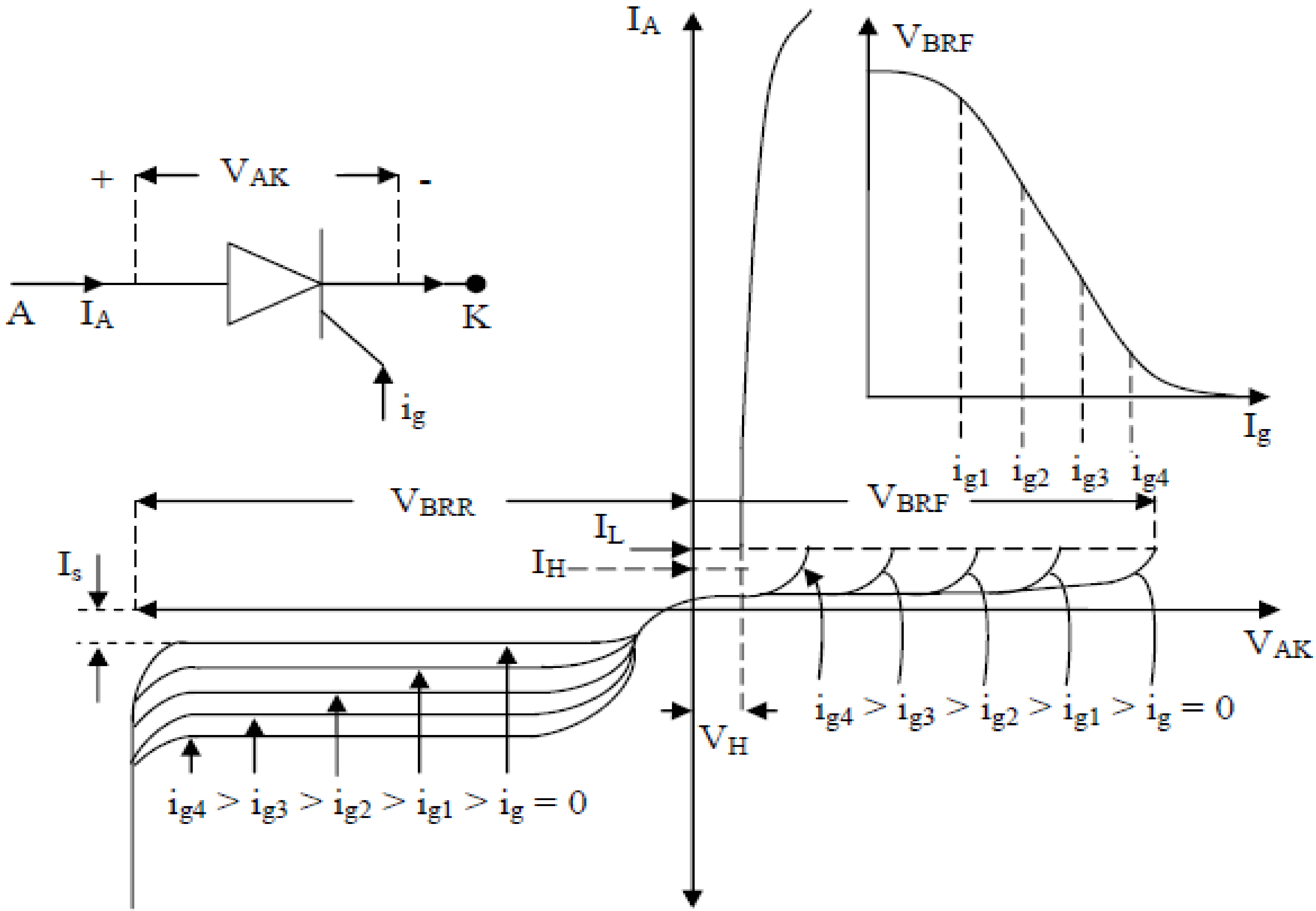
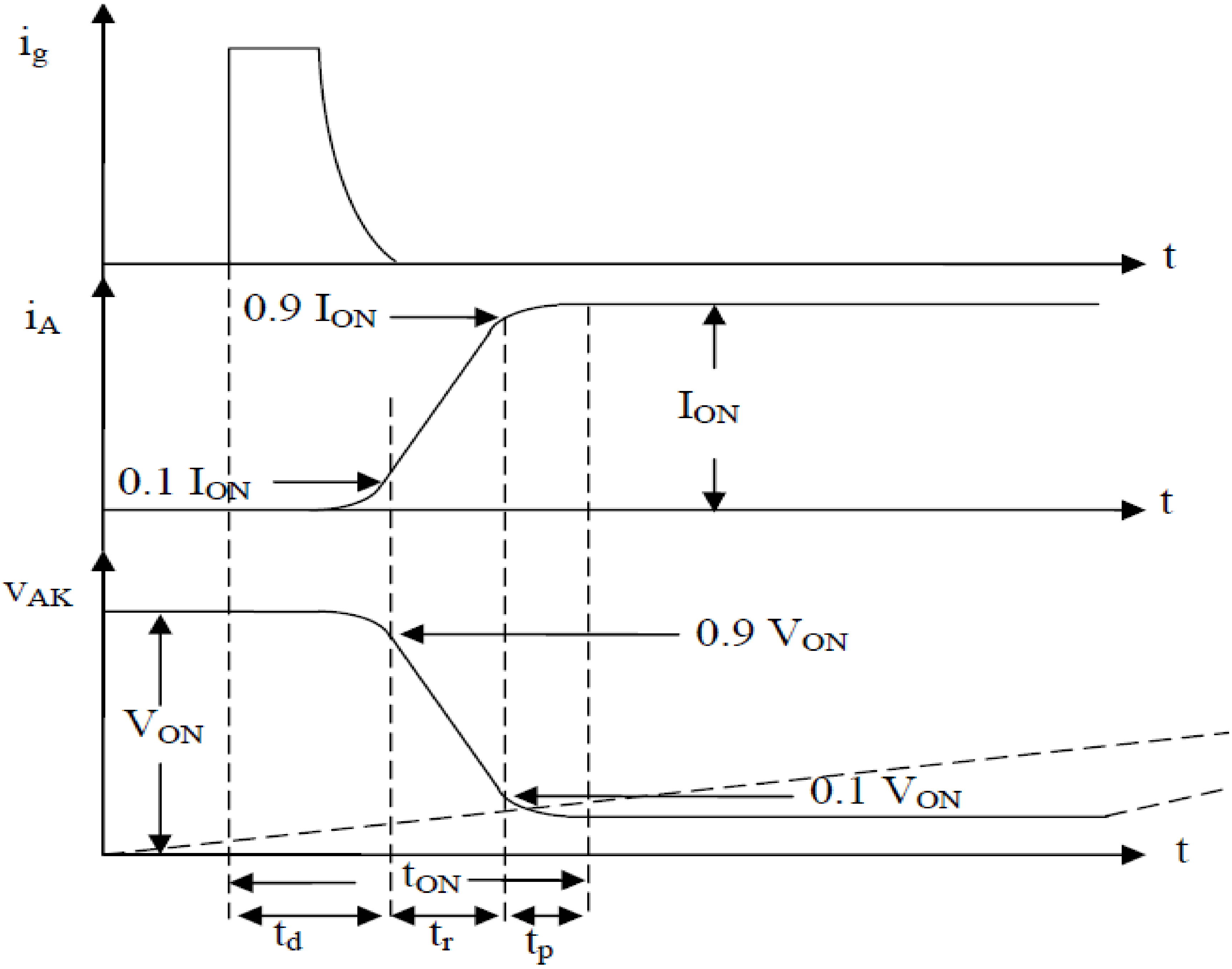


Fig. 4.3: Static output characteristics of a Thyristor

Hold and Latch!

- The MINIMUM magnitude of anode current **beyond** which a forward biased thyristor will remain ON, even after gate signal removal, is termed latching current. It is defined only for Turn ON process.
- The MAXIMUM magnitude of anode current **beneath** which a thyristor will start to turn OFF, in the absence of gate signal, is termed holding current. It is defined only for Turn OFF process.
- Latching current is greater than holding current.

Thyristor Turn On Characteristics



i_g
t
i_A
$0.9\ I_{ON}$
I_{ON}
$0.1\ I_{ON}$
t
v_{AK}
$0.9\ V_{ON}$
V_{ON}
$0.1\ V_{ON}$
t
t_{ON}
t_d
t_r
t_p

Thyristor Turn Off Characteristics

Same as Diode